

Lesson 01

Understanding Server Administration Concepts



Topic 1A

Understand Server Administration Concepts

Exam Objectives Covered

- This topic provides background information about the role of server administrators and does not cover a specific exam objective

System Administrator Job Roles and Responsibilities

- Common Sysadmin Responsibilities

System Administrator Job Roles and Responsibilities

- Common Sysadmin Responsibilities
 - Installing, configuring, and managing hardware, apps, and network components
 - Managing the lifecycle
 - Monitoring performance
 - Capacity planning
 - Access control
 - Troubleshooting
 - Liaising with other IT teams
 - Resolving escalated support tickets
 - Managing disparate technologies and network services
 - Providing documentation
 - Improving communication and presentation skills
 - Advising on and implementing security policies

Servers Versus Workstations

Workstation Characteristics

- Assigned to end user
- No redundant hardware
- No high-performance hardware
- Client OS optimized for single user
- Apps optimized for a single user
- Elegant GUI
- Desktop or laptop form factor

Server Characteristics

- Secured in a server room
- Redundant hardware
- High-performance hardware
- Server OS optimized for many users
- Apps optimized for multiple users
- Limited or no GUI
- Rack-mounted form factor

Servers Versus Workstations (slide 2)

Workstation Operating Systems

- Microsoft Windows 10
- Ubuntu Desktop Linux
- MacOS

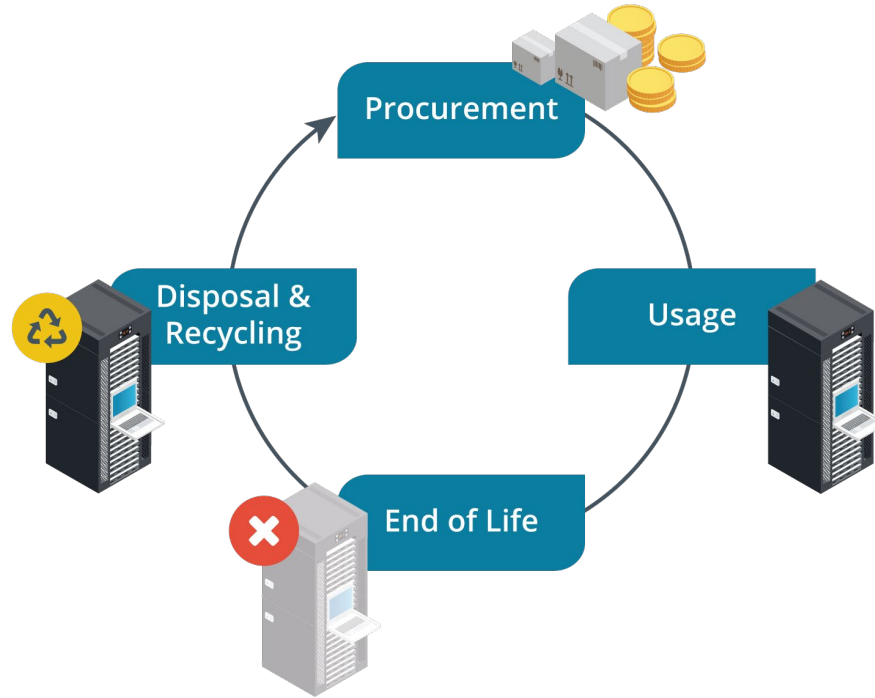
Server Operating Systems

- Microsoft Windows Server 2019
- Red Hat Enterprise Linux 8 (RHEL 8)

Servers Versus Workstations (slide 3)

- Workstation and Server Security

Server Lifecycle



Server lifecycle illustration. (Images © 123RF.com)

The Four Major Subsystems

- Processor
- Memory
- Storage
- Network

Server OSs—Linux and Windows

- Linux
 - Red Hat Linux Derivatives
 - RHEL
 - Fedora
 - CentOS
 - Debian Linux Derivatives
 - Debian Linux
 - Ubuntu
 - Kali
- Windows Server

Review Activity: Server Administration Concepts



Review Activity

Topic 1B

Understand Troubleshooting Methods

Exam Objectives Covered

- 4.1 Explain the troubleshooting theory and methodology

Troubleshooting Methodology

- Identify problem
- Determine scope
- Establish theory of probable cause/question the obvious
- Test theory to determine cause
- Establish action plan
- Implement solution or escalate
- Verify full system functionality
- Implement preventative measures
- Perform root cause analysis
- Document findings, actions, and outcomes

Identify the Problem and Determine the Scope

- Identify problem
- Determine scope

Establish and Test a Theory of Probable Cause

- Establish theory of probable cause/question the obvious
- Test the theory to determine the cause

Establish and Implement a Plan of Action

- Establish a plan of action
- Implement the solution or escalate

Verify, Prevent, Analyze, and Document

- Verify full system functionality
- Implement preventative measures
- Perform a root cause analysis
- Document findings, actions, and outcomes throughout the process

Guidelines for Troubleshooting Methods

- Troubleshooting is a key skill for sysadmins
- A troubleshooting methodology is a more efficient and accurate process
- The methodology could change depending on the situation

Review Activity: Discuss a Troubleshooting Methodology



Review Activity

Topic 1C

Manage Licenses

Exam Objectives Covered

- 2.8 Explain licensing concepts

Licenses Versus Maintenance and Support

- License types
 - FOSS
 - Proprietary
- Maintenance and Support Plans

Licensing Models

- Subscription
- Volume licensing
- Per-instance
- Per-concurrent-user
- Per-seat license
- Per-server
- Per-socket
- Per-core
- Site-based
- Node-locked
- Physical vs. virtual

Licensing Models (slide 2)

- Electronic signatures
- Version compatibility

Guidelines for Managing Licenses

- FOSS licensing allows for modification of source code and distribution of modified software
- Proprietary licensing does not allow source code modification
- Choose a method to track licenses to ensure compliance
- Understand licensing methods
- A vendor may offer different licensing models
- Know whether your license includes maintenance and support
- Cloud-based services often use a subscription-based model

Review Activity: Licenses



Review Activity

Lesson 1

Summary

